

# PRD: AI-Powered Road Accident Risk Prediction System (India)

## 1. Project Overview

### Objective

Build an end-to-end Machine Learning application that predicts road accident risk and severity in India, using a record-level Kaggle accident dataset as the training source. The project should demonstrate a complete ML lifecycle including data engineering, machine learning, API development, frontend development, deployment, and basic MLOps practices.

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## 2. About the Data

### Dataset

**India Road Accident Dataset - Predictive Analysis** (Kaggle, by khushikyad001) Source: <https://www.kaggle.com/datasets/khushikyad001/india-road-accident-dataset-predictive-analysis>

This is a record-level (accident-by-accident) dataset covering the years 2018–2023, intended specifically for predictive modeling and risk assessment. Each row represents a single accident with the following attributes:

- State and city of occurrence
- Year, month, day of week, and exact time (HH:MM) of the accident
- Accident severity: Fatal / Serious / Minor
- Number of vehicles involved and vehicle type (Car, Truck, Two-Wheeler, etc.)
- Driver age, gender, and license status
- Weather condition, lighting condition, road type, and speed limit at the accident location
- Alcohol involvement (drunk driving flag)

This dataset supplies the record-level features needed to train a per-accident risk/severity model.

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## 3. Problem Statement

Road accident data is often analyzed only in hindsight through static reports.

The objective is to build a unified system that:

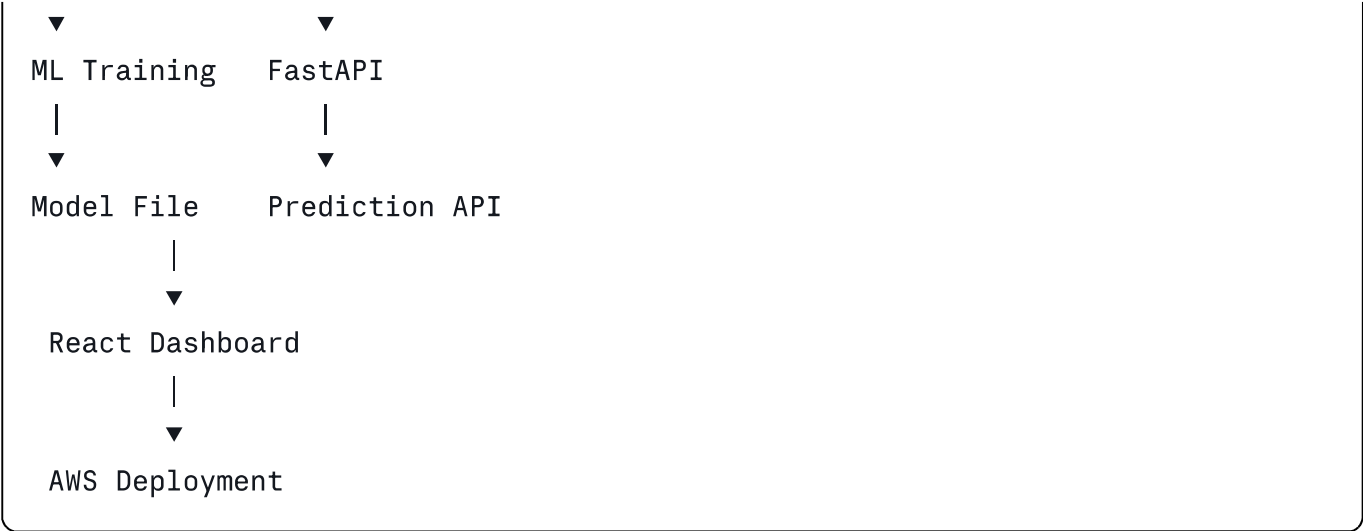
- Cleans and prepares accident-level data
- Generates meaningful features
- Trains ML models
- Predicts accident risk/severity
- Provides interactive visualizations

## 4. Technology Stack

| Layer         | Choice                                                    |
|---------------|-----------------------------------------------------------|
| Frontend      | React (or HTML/CSS/JS)                                    |
| Backend       | FastAPI                                                   |
| Database      | PostgreSQL (or MySQL)                                     |
| ML            | Pandas, NumPy, Scikit-learn, XGBoost, LightGBM (optional) |
| Visualization | Plotly, Chart.js, Recharts                                |
| Deployment    | Docker, AWS EC2, Nginx, GitHub Actions                    |

## 5. System Architecture





## 6. Data Sources

**India Road Accident Dataset – Predictive Analysis** (Kaggle, khushikyad001), 2018–2023:  
(state, city, year, month, day\_of\_week, time, severity, num\_vehicles, vehicle\_type, driver\_age, driver\_gender, license\_status, weather, lighting, road\_type, speed\_limit, alcohol\_involved)

## 7. ML Pipeline

Collect Data → Clean Data → EDA → Feature Engineering → Train/Test Split → Train Models → Evaluate → Save Best Model → Deploy

**Target variable:** Accident severity (Fatal / Serious / Minor) — framed as a classification problem.

**Models to implement and compare:** Logistic Regression, Random Forest, XGBoost. Select best model based on evaluation metrics.

## 8. API Endpoints

| Method | Endpoint | Purpose                                                                                                                                       |
|--------|----------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| POST   | /upload  | Upload dataset                                                                                                                                |
| POST   | /predict | Return prediction based on user inputs (state, year, month, day of week, time, vehicle type, driver age/gender, weather, lighting, road type, |

| Method | Endpoint    | Purpose                                         |
|--------|-------------|-------------------------------------------------|
|        |             | speed limit, alcohol involvement)               |
| GET    | /statistics | Return dashboard statistics                     |
| GET    | /model-info | Return model information and evaluation metrics |

## 9. Database Schema

**accidents** (record-level, from Kaggle dataset)

```
id, state, city, year, month, day_of_week, time,
severity, num_vehicles, vehicle_type,
driver_age, driver_gender, license_status,
weather, lighting, road_type, speed_limit,
alcohol_involved
```

**predictions**

```
id, state, prediction, created_at
```

## 10. Folder Structure

```
road-accident-ml/
|
├─ data/
├─ notebooks/
├─ backend/
|   └─ api/
|   └─ models/
|   └─ services/
|   └─ database/
├─ frontend/
├─ ml/
|   └─ training/
|   └─ preprocessing/
|   └─ inference/
```

- └─ docker/
- └─ README.md

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## 11. MLOps Workflow

GitHub → GitHub Actions → Run Tests →  
Train Model (Manual/Triggered) → Store Model Artifact →  
Docker Build → Deploy to AWS EC2 → Serve Predictions via FastAPI